

# Digital Twin System for Reflector Antenna Structure Performance Evaluation Based on Surrogate Model

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**Abstract**—The digital twin system of the reflector antenna based on the surrogate model is designed in this paper to achieve real-time evaluation of its structural performance during operation. Firstly, the finite element model of the reflector antenna is established, and mechanical performance data samples of the antenna structure are obtained. Secondly, using surrogate model technology, a high-precision Gaussian Process mathematical model is fitted to enable rapid and accurate prediction of the antenna structure's performance. Finally, data communication technology is utilized to connect the physical space of the antenna with its twin space, enabling visualization of its structure's performance and interaction with motion attitude based on the Unity engine. Through testing, it has been demonstrated that this digital twin system can achieve real-time monitoring of high precision and efficiency for the structural performance of the antenna.

**Keywords**—reflector antenna, surrogate model, digital twin, Gaussian Process, real-time, high precision

## I. INTRODUCTION

The reflector antenna is an antenna that focuses and guides electromagnetic waves through a metal reflector to form a high-gain, narrow-beam antenna [1]. It is widely used in radar detection, satellite communications, radio astronomy, and other fields. During operation, external factors such as gravity, wind, and temperature can cause structural deformation of the reflector antenna, which can significantly impact its electrical performance. Therefore, real-time monitoring of the structural performance status of the reflector antenna becomes crucial.

Most of the current methods for antenna operation and maintenance adopt modular distributed management, with each system operating independently. Achieving global monitoring and management is difficult, resulting in weak interactive performance and poor visualization ability. The emergence of digital twin (DT) technology provides an innovative intelligent management method for the field of antenna operation and maintenance. DT technology utilizes digital methods to construct a high-fidelity virtual model for physical entities. By relying on Internet of Things communication technology, the virtual model can synchronously reflect the operational status of physical entities through data interaction and mapping, while also

offering simulation, monitoring, and early warning functions [2]. DT technology is currently being applied in the field of antennas: [3] constructed the DT model for the FAST reflector unit and applied it to the digital protection of the antenna. [4] designed the DT system for the FAST feed cabin, which improved the automation management of the antenna. [5] established a DT system for the antenna array, enabling assembly and regulation. Although these DT systems allow real-time monitoring, there are still deficiencies in dynamic visualization and rapid prediction of antenna structural performance.

In this paper, a DT system is designed for monitoring the structural performance of reflector antennas by combining finite element technology, surrogate model technology, and DT technology. The real-time high-precision prediction of the reflector antenna's structural performance is achieved through the construction of a data-driven GP surrogate model.

## II. DT SYSTEM FRAMEWORK

The technical framework of the DT system of the reflector antenna is shown in Fig. 1. The entire framework mainly consists of two parts: physical space and twin space. The physical space includes antenna entities, acceleration sensors, STM32, and other hardware devices. The digital space comprises a virtual model, behavior model, mechanism information, and prediction surrogate model based on finite element simulation data. Data interaction between the physical space and the twin space is achieved through Socket communication and serial communication. Unity client serves as the carrier of twin space to realize real-time prediction and high-fidelity visualization of the structural performance of the reflector antenna.

## III. SURROGATE MODEL CONSTRUCTION

### A. Data Acquisition

In this paper, the primary reflector of the antenna is taken as an example. The finite element model is established based on the Ansys Workbench. The back frame adopts solid element, the panel adopts shell element, and tetrahedral elements are used for meshing. Under ideal working conditions, a standard gravity load is applied to calculate the structural deformation of the main reflector at different pitch

angles through finite element analysis. Fig. 2 shows the deformation cloud diagram of the main reflector at pitch angles of 90° and 45°.

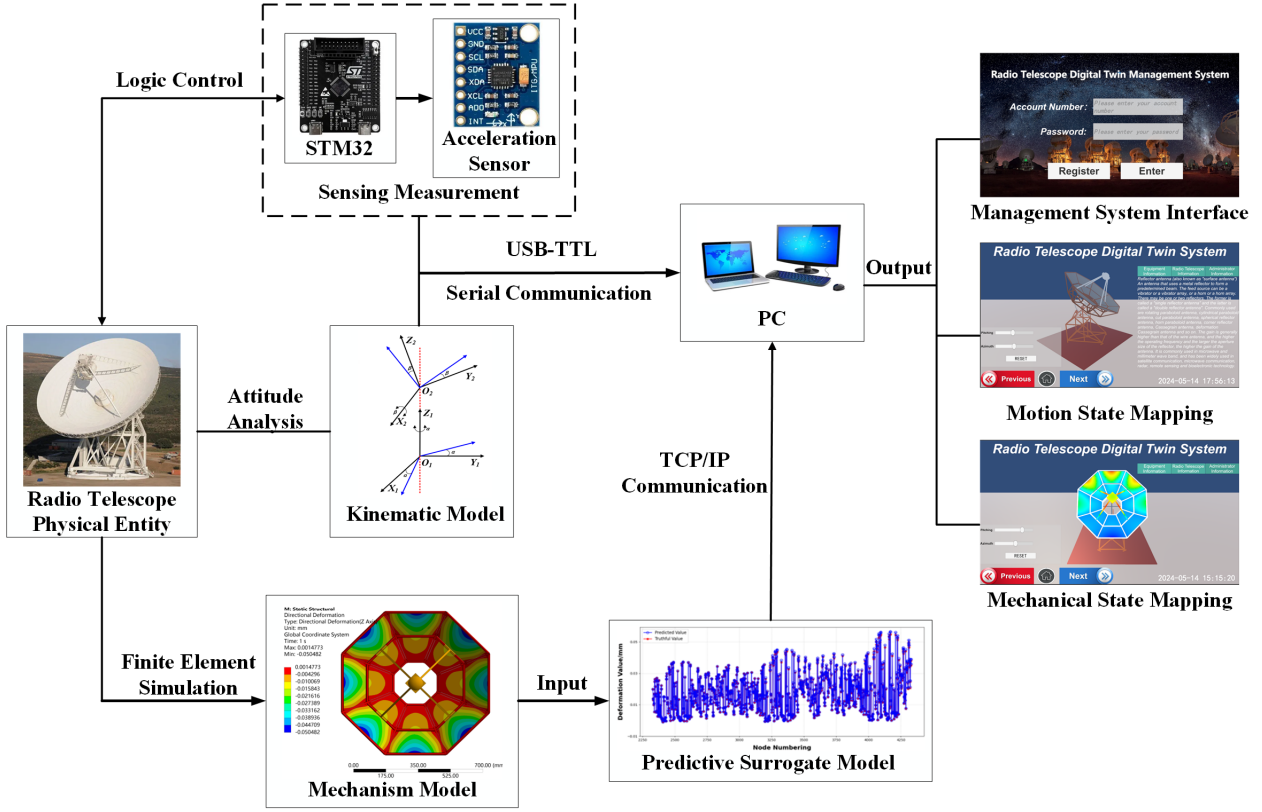


Fig. 1. Technical framework of DT system for reflector antenna.

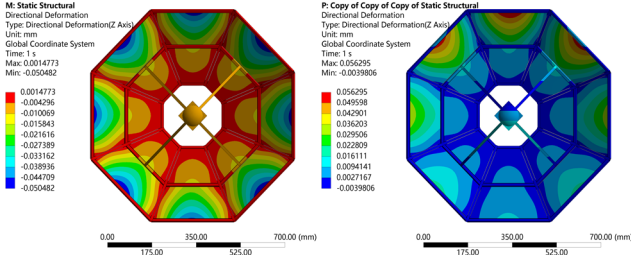


Fig. 2. The deformation results of the main reflector at pitch angles of 90° (left) and 45° (right).

### B. Gaussian Process

Gaussian Process (GP) has a good simulation effect and generalization ability in dealing with small samples and high-dimensional complex problems [6]. In this paper, the GP surrogate model is used to predict the structural performance of the antenna main reflector in real-time. The distribution characteristics of GP are determined by its mean function  $m(x)$  and covariance function  $c(x, x')$ , and its general form is expressed as [7,8]:

$$y \sim gp(m(x), c(x, x')) \quad (1)$$

$$\begin{cases} m(x) = E[f(x)] \\ c(x, x') = E[(f(x) - m(x))(f(x') - m(x')))] \end{cases}$$

For a given data set  $D = \{(x_i, y_i)\}_{i=1}^n$ , where  $x_i \in R^d$  is the input matrix and  $y_i \in R$  is the output matrix. Assuming that  $\{f(x_1), f(x_2), \dots, f(x_{n-1}), f(x_n)\}$  is a set of random variables following a joint Gaussian distribution, the expression of the GP regression distribution is as follows:

$$f(\mathbf{x}) \sim N(\boldsymbol{\mu}, \mathbf{C}) \quad (2)$$

In the formula, the mean function  $m$  in this paper is chosen as a constant form denoted by  $\boldsymbol{\mu}$ ; the covariance function selects the square exponential function, and its matrix form is denoted by  $\mathbf{C}$ . The expression is as follows:

$$\mathbf{C}(X, X) = \eta^2 \exp \left[ -\frac{1}{2} \sum_{k=1}^n \left( \frac{X_i^k - X_j^k}{\omega_k} \right)^2 \right] \quad (3)$$

There is a new point  $x^*$  to be measured, which is assumed by the GP. The predicted values  $y^*$  and  $f(\mathbf{x})$  also follow the Gaussian distribution, that is :

$$\begin{bmatrix} \mathbf{Y} \\ y^* \end{bmatrix} \sim N \left( \begin{bmatrix} \boldsymbol{\mu}(\mathbf{X}) \\ \boldsymbol{\mu}(\mathbf{x}^*) \end{bmatrix}, \begin{bmatrix} \mathbf{C}(\mathbf{X}, \mathbf{X}) & \mathbf{C}(\mathbf{X}, \mathbf{x}^*) \\ \mathbf{C}(\mathbf{x}^*, \mathbf{X}) & \mathbf{C}(\mathbf{x}^*, \mathbf{x}^*) \end{bmatrix} \right) \quad (4)$$

From this, the main prediction equation of GP regression can be obtained, and its expression is :

$$f^* | X^*, X, Y \sim N\left(\bar{f}^*, \text{cov}(\bar{f}^*)\right) \quad (5)$$

Where  $\bar{f}^*$  represents the mean value of the predicted value in the GP regression model, and  $\text{cov}(\bar{f}^*)$  represents the variance of the predicted value. For a sample data set from a finite element model with  $k$  nodes, it is necessary to establish  $k$  GP surrogate models  $\bar{f}_{(k)}^*$  to predict its structural performance.

### C. Comparison of Results

In this paper, seven sets of working condition data for the main reflector of the reflector antenna at different pitch angles ( $\beta = 5^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ$ ) were selected for training the GP surrogate model. Four random pitch angles were chosen as test conditions to evaluate the prediction accuracy using root mean square error (RMSE) and coefficient of determination ( $R^2$ ). The formulas are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

In the formula,  $n$  represents the number of nodes in the finite element model;  $y_i$  denotes the simulation value obtained from finite element analysis;  $\bar{y}$  represents the mean value of finite element results; and  $\hat{y}_i$  signifies the predicted value of the GP model.

Table I shows the comparison of prediction accuracy under four random working conditions, and Fig. 3 shows the comparison of predicted values and simulation values corresponding to 2000 consecutive nodes under  $\beta=38.6^\circ$ . It can be seen that compared with the finite element analysis, the RMSE of the GP surrogate model is maintained at a low level, and the  $R^2$  are all greater than 0.99, showing good prediction accuracy.

TABLE I. PREDICTION ACCURACY OF GP SURROGATE MODEL UNDER TEST CONDITIONS

| Test Condition | RMSE(mm) | $R^2$ |
|----------------|----------|-------|
| 80.5°          | 5.9E-4   | 0.996 |
| 61.4°          | 4.72E-5  | 0.999 |
| 38.6°          | 4.37E-4  | 0.998 |
| 21.2°          | 4.41E-4  | 0.998 |

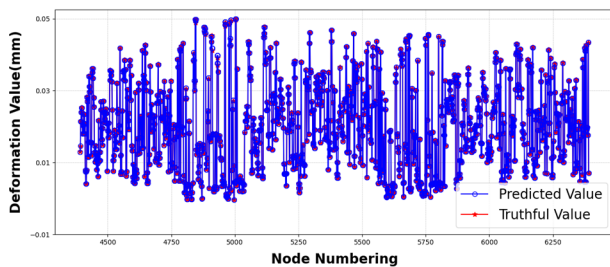


Fig. 3. Comparison of predicted and simulated values ( $\beta = 38.6^\circ$ ).

Secondly, while meeting the accuracy requirements, the calculation time of the two models is compared. Table II presents a comparison of the time taken for a single solution. It can be observed that in comparison to the finite element model, the GP surrogate model improves calculation efficiency by approximately 98.38%, enabling rapid and accurate prediction of structural deformation in the main reflector of the reflector antenna.

TABLE II. COMPARISON OF CALCULATION TIME LENGTH

| Model Category       | Computation Time |
|----------------------|------------------|
| finite element model | 21 s             |
| GP delegation model  | 0.34 s           |

### IV. DATA VISUALIZATION

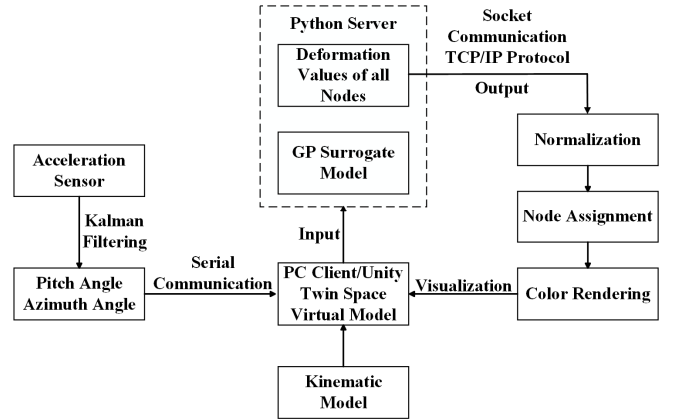


Fig. 4. Data visualization technology route.

The technical route of data visualization is shown in Fig. 4. This paper utilizes a six-axis acceleration sensor to obtain the attitude angle of the reflector antenna. The Kalman filter algorithm is employed to process the sensing data, which is then transmitted to the Unity client through serial communication. By combining it with the motion model of the reflector antenna, real-time mapping of attitude from physical space to twin space is achieved. Additionally, Python acts as a server and establishes a Socket interface with the client. Based on TCP/IP communication protocol, received real-time angle data serves as input for the GP surrogate model, which outputs deformation values for all grid nodes after solving. Upon receiving deformation data, Unity normalizes it and assigns model grid nodes using HSV color space [9]. Finally, a dynamic display of antenna structure deformation in cloud image form is realized. The rendered effect can be seen in Fig. 5.

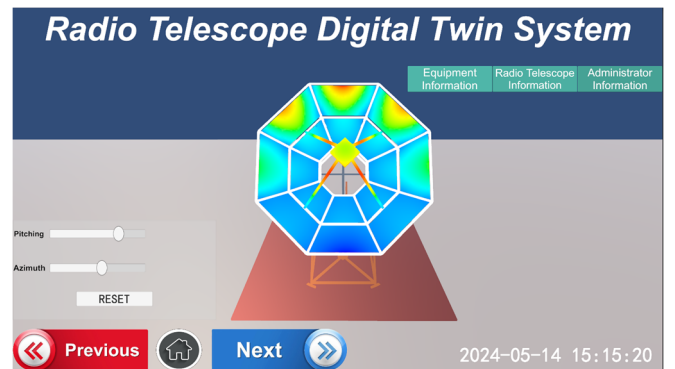


Fig. 5. Visual renderings.

## V. CONCLUSION

In this paper, a DT system is designed to visualize the structural performance of reflector antennas. Based on finite element simulation data, a high-precision GP surrogate model is constructed. It has been verified that the prediction model has a small RMSE, a coefficient of determination greater than 0.99, and an improved computational efficiency of about 98.38%. This realization enables high-precision and fast prediction of the structural mechanical state of the reflector antenna. Finally, real-time mapping of motion state and real-time rendering of structural deformation is achieved through the Unity client to complete the construction of the reflector antenna DT system.

## ACKNOWLEDGMENT

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